Left and right production matrices and total positivity of different types

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Started on December 19, 2025. Last updated onAugust 1, 2026 **need to put final date!!!**

Abstract

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Key Words:

Mathematics Subject Classification (MSC 2010) codes:

Contents

1 Introduction

A (finite or infinite) matrix of real numbers is called ***totally positive*** (TP) if all its minors are nonnegative. A matrix of polynomials with real coefficients is called ***coefficientwise totally positive*** if all its minors are polynomials with nonnegative coefficients. Let $A = (a_{n,k})_{n,k\geq 0}$ be a lower-triangular matrix of nonnegative real numbers. In [8], some of us noticed that for such a matrix, we may always ask the following four questions:

(a) Is A totally positive?

(b) Is the lower-triangular matrix $\overleftarrow{A} \stackrel{\text{def}}{=} (a_{n,n-k})_{n,k\geq 0}$, obtained by reversing the rows of A, totally positive? (Here $a_{n,n-k} \stackrel{\text{def}}{=} 0$ when $n < k$.)

(c) Let $A_n(x) = \sum_{k=0}^{n} a_{n,k} x^k$ denote the row-generating polynomial of the n-th row of A. Are the polynomials $A_n(x)$ negative-real-rooted?[1] This is equivalent [13, pp. 395, 399] to asking if the Toeplitz matrix of the n-th row sequence $(a_{n,k})_{k\geq 0}$ — that is, the matrix $(a_{n,i-j})_{i,j\geq 0}$ with $a_{n,k} \stackrel{\text{def}}{=} 0$ for $k < 0$ — is totally positive. In this situation we say that the n-th row sequence $(a_{n,k})_{k\geq 0}$ is ***Toeplitz-totally positive***.

(d) Is the Hankel matrix of the polynomial sequence $(A_n(x))_{n\geq 0}$ — that is, the matrix $(A_{i+j}(x))_{i,j\geq 0}$ — coefficientwise totally positive in the variable x? In this situation we say that the sequence $(A_n(x))_{n\geq 0}$ is ***coefficientwise Hankel-totally positive***.

For several important combinatorial triangles, the answers to all four of these questions have been proven to be yes: these include the matrix of binomial coefficients, the matrix of Stirling cycle numbers, the matrix of Stirling subset numbers; see [8] for details. Of these, the total positivity of the reversal of the Stirling subset numbers was only recently shown by us few years ago [4]. Next consider $A = \left(\left\langle {n \atop k} \right\rangle\right)_{n,k\geq 0}$, the matrix of Eulerian numbers, which count the permutations on $[n]$ with k descents (or k excedances). In this case, the answers to questions (c) and (d) is known to be true [12, 23]. However, here the total positivity of the matrix A is equivalent to the total positivity of the matrix $\overleftarrow{A}$, and even though this was conjectured by Brenti in 1996 [3] the question is still open even after three decades.

A sufficient condition that has often been used to prove questions (a),(d) true simultaneously is the *method of production matrices* see for example [**?**]. On the other hand, one of us along with Fu and Ruan [5], provided a sufficient condition that implies the truth of questions (a), (b), (c) simultaneously: the existence of a *totally positive left-production matrix.*

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[1] A polynomial with real coefficients is said to be ***real-rooted*** if it is either identically zero or all its (complex) zeros are real; it is said to be ***negative-real-rooted*** if it is either identically zero or all its (complex) zeros lie in $(-\infty, 0]$. Since we are here assuming that all the matrix entries $a_{n,k}$ are nonnegative, clearly $A_n(x)$ cannot have any *positive* real zeros unless it is identically zero. So negative-real-rootedness and real-rootedness are equivalent here.

The most general setting to consider the questions (a),(b),(c),(d) is to take a partially ordered commutative ring R and let the entries of the matrix A be $a_{nk} \in R$. The notions of total positivity, Hankel-total positivity and Toeplitz-total positivity are then defined in the obvious way. The main result of this paper is Theorem 2.3 which generalises [5, Theorem 1.3], which is in the setting when $R = \mathbb{R}$, to the setting of a partially ordered commutative ring; see Section 2.2. We will then write out the relation between the two notions of production matrices and left-production matrices in Section 2.3. In this paper, we will refer to production matrices as *right-production matrices.*

We remark that the proof of [5, Theorem 1.3], uses the fact that any invertible lower-triangular matrix with real entries has a planar digraph representation ([5, Theorem 2.5]); this fact itself relies on a bidiagonal factorisation of such a matrix, see [13, Theorem 6.10]. However, such a bidiagonal factorisation may not exist if R is not a field. Our proof of Theorem 2.3 is algebraic and works for any general partially ordered commutative ring R.

After developing the general theory, we will write out some applications of Theorem 2.3. Our first main application (Section 4) here is an algebraic proof of the total positivity of the *ace*-matrix in [4, Theorem 1.5] that we had promised five years ago. We will also show that the row-generating Toeplitz matrix is totally positive.

Our second application will be some results concerning the multivariate Laguerre coefficient matrix [7], see Section 5.1. Our final application (Section 6) is a proof of the total positivity of the matrix of linearly ordered phylogenetic trees [8, Conj. 4.17].

Finally, after raising questions (a),(b),(c),(d), one wonders if these have any relations or implications among each other. We will visit this in Section 7 where we construct an example for each of the $2^4 = 16$ different truth table values to the four questions. This will show that there is no relation among these four questions in general.

2 Left production matrices

2.1 Definition of left production matrices

Let Q be a row-finite or column-finite matrix. Define the following matrix $\widehat{A}$

$$\widehat{A} \;=\; Q \left[\begin{array}{c|c} I_1 & \mathbf{0} \\ \hline \mathbf{0} & Q \end{array}\right] \left[\begin{array}{c|c} I_2 & \mathbf{0} \\ \hline \mathbf{0} & Q \end{array}\right] \left[\begin{array}{c|c} I_3 & \mathbf{0} \\ \hline \mathbf{0} & Q \end{array}\right] \cdots \quad . \tag{2.1}$$

Note that (2.1) can equivalently be rewritten as

$$\widehat{A} \;=\; Q \left[\begin{array}{c|c} I_1 & \mathbf{0} \\ \hline \mathbf{0} & \widehat{A} \end{array}\right] \tag{2.2}$$

since (2.1) imply (2.2) by substitution on the right-hand side, while (2.2) imply (2.1) by iteration. We call Q the ***left-production matrix***, and $\widehat{A}$ the ***left-output ma-***

trix. We may rewrite (2.2) as the recurrence

$$\widehat{a}_{n0} = q_{n0} \tag{2.3a}$$

$$\widehat{a}_{n,k+1} = \sum_{i=0}^{\infty} q_{n,i+1}\widehat{a}_{i,k} \quad \text{for } n,k \ge 0 \tag{2.3b}$$

Fully carrying out the recurrence, we have

$$\widehat{a}_{nk} = \sum_{i_1,\ldots,i_k} q_{n,i_1+1} q_{i_1,i_2+1} \cdots q_{i_{k-1},i_k+1} q_{i_k,0} \ . \tag{2.4}$$

If we further assume that Q is lower-Hessenberg, we get the recurrence

$$\widehat{a}_{n,k+1} = \sum_{i=0}^{n} q_{n,i+1}\widehat{a}_{i,k} \quad \text{for } n,k \ge 0 \tag{2.5}$$

with initial condition $\widehat{a}_{n0} = q_{n0}$. In this case, the matrix $\widehat{A}$ is a dense matrix with zeroth row given by $\widehat{a}_{0k} = q_{00}\, q_{01}^k$.

Finally, if we assume that Q is lower-triangular, than the matrix A is also lower triangular and we get the recurrence

$$\widehat{a}_{n+1,k+1} = \sum_{i=0}^{n} q_{n+1,i+1}\widehat{a}_{i,k} \quad \text{for } n,k \ge 0 \tag{2.6}$$

with initial condition $\widehat{a}_{n0} = q_{n0}$. The diagonal elements of $\widehat{A}$ are partial products of those of Q:

$$\widehat{a}_{nn} = \prod_{i=0}^{n} q_{ii} \ . \tag{2.7}$$

In particular, $\widehat{A}$ is invertible $\iff$ the diagonal elements of $\widehat{A}$ are invertible $\iff$ the diagonal elements of Q are invertible $\iff$ Q is invertible. And in this case, Q can be recovered from $\widehat{A}$ by

$$Q = \widehat{A} \cdot \left[\begin{array}{c|c} I_1 & \mathbf{0} \\ \hline \mathbf{0} & \widehat{A}^{-1} \end{array}\right] = \widehat{A}\left(\mathbf{e}_{00} + \Delta^{\mathrm{T}}\widehat{A}\Delta\right)^{-1} \tag{2.8}$$

where Δ is the matrix with 1 on the superdiagonal and 0 elsewhere.

GIVE SOME HISTORY!!!!!!

We now write the following proposition.

Proposition 2.1. *Let Q be a row-finite matrix with entries in a commutative ring R, with left-output matrix $\widehat{A}$; and let B be an invertible row-finite matrix such that B^{-1} is also row-finite. Then the matrix $B\widehat{A}$ is the left-output matrix of the following matrix*

$$B\,Q \left[\begin{array}{c|c} I_1 & \mathbf{0} \\ \hline \mathbf{0} & B^{-1} \end{array}\right] . \tag{2.9}$$

Proof. Let Q_2 be the matrix in (2.9). We write out (2.1) with the matrix Q_2 and see that it gives us BA. □

2.1.1 Combinatorial interpretation

Let $Q = (q_{ij})_{i,j\geq 0}$ be a lower-triangular matrix and let $\widehat{A} = (\widehat{a}_{nk})_{n,k\geq 0}$ be its left-output matrix. We now provide a combinatorial interpration of the matrix entries of $\widehat{A}$.

Consider the alphabet $\mathbb{N} \cup \{|\}$ and set w_n to be the word $w_n = 0\,1\,2\,\cdots\,n$ on this alphabet. Let $\mathbb{B}_n$ be the set of all words obtained by inserting bars "|" between any two consecutive letters of w_n. For example $\mathbb{B}_2 = \{0\,1\,2, 0|1\,2, 0\,1|2, 0|1|2\}$. Clearly, the set $\mathbb{B}_n$ is in bijection with subsets of the set $[n]$ and therefore has cardinality 2^n: a subset $S \subseteq [n]$ encodes the letters occuring before bars in the word $w \in \mathbb{B}_n$. We use $\mathbb{B}_{n,k} \subseteq \mathbb{B}$ to denote the subcollection of words each containing k bars.

To each finite interval $[i,j] \subset \mathbb{N}$, we assign the weight $\mathrm{wt}([i,j]) = q_{ji}$. The weight of a word $w \in \mathbb{B}_{n,k}$, denoted by $\mathrm{wt}(w)$ is the product of the weights of the $(k+1)$ subintervals of $[0,n]$ separated by the bars in w. For example, the words in $\mathbb{B}_2$ have the following weights:

$$\mathrm{wt}(0\,1\,2) \;=\; q_{2,0} \quad \mathrm{wt}(0|1\,2) \;=\; q_{0,0}q_{2,1} \quad \mathrm{wt}(0\,1|2) \;=\; q_{1,0}q_{2,2} \quad \mathrm{wt}(0|1|2) \;=\; q_{0,0}q_{1,1}q_{2,2}\,.$$

Finally, we define $\mathrm{wt}(\mathbb{B}_{n,k}) := \sum_{w\in\mathbb{B}_{n,k}} \mathrm{wt}(w)$. For example, $\mathrm{wt}(\mathbb{B}_{2,1}) = q_{0,0}q_{2,1} + q_{1,0}q_{2,2}$.

We are now ready to state our combinatorial interpretation of the left-output matrix $\widehat{A}$.

Proposition 2.2. *For a lower triangular left-production matrix $Q = (q_{ij})_{i,j\geq 0}$ with corresponding left-output matrix $\widehat{A} = (\widehat{a}_{nk})_{n,k\geq 0}$, the entry $\widehat{a}_{n,k}$, is given by the identity:*

$$\widehat{a}_{n,k} \;=\; \mathrm{wt}(\mathbb{B}_{n,k})\,. \tag{2.10}$$

Proof. First notice that $\mathrm{wt}(\mathbb{B}_{n,0}) = (\{w_n\}) = q_{n0}$; thus the initial condition is satisfied. It is easy to see that the quantities $\mathrm{wt}(\mathbb{B}_{n,k})$ satisfy the recurrence (2.6). We omit the details. $\square$

2.2 Left production matrices and total positivity

We now state the main result of this paper.

Theorem 2.3. *Let $Q = (q_{ij})_{i,j\geq 0}$ be a row-finite or a column-finite left-production matrix with elements in a partially ordered commutative ring R, let $\widehat{A} = (\widehat{a}_{nk})_{n,k\geq 0}$ be its left-output matrix. Let us assume that Q is totally positive of order r. We then have*

- *(a) The matrix $\widehat{A}$ is also totally positive of order r.*
- *(b) If Q is lower triangular (so that the matrix $\widehat{A}$ is lower triangular), then the reversal $\overleftarrow{\widehat{A}}$ exists and it is totally positive of order r.*
- *(c) Let Q be either lower-triangular, or more generally, lower-Hessenberg. For any fixed row $n \geq 0$, the infinite lower triangular Toeplitz matrix $T_\infty((\widehat{a}_{n,k})_{k\geq 0}) = (\widehat{a}_{n,i-j})_{i,j\geq 0}$ is Toeplitz-totally positive of order r.*

This is a striking result. **SAY WHY???????**

Remark. Theorem 2.3 was proved by one of us in the setting when $R = \mathbb{R}$, and for total positivity of order ∞ and also Q was assumed to be lower triangular; see [5, Theorem 1.3]. The proof in this case is via the existence of a planar network representation of the matrix Q ([5, Theorem 2.5]) which itself relies on a bidiagonal factorisation of a totally positive matrix with real entries, see [13, Theorem 6.10].

However, a bidiagonal factorisation may not exist if R is not a field. Our proof of here is algebraic and works for a general partially ordered commutative ring. ■

In preparation for the proof of Theorem 2.3, let us introduce some useful notation. For any infinite matrix $X = (x_{ij})_{i,j\geq 0}$ and any $M, N \in \mathbb{N}\cup\{\infty\}$, let us write $X_{M\times N} = (x_{ij})_{0\leq i\leq M-1, 0\leq j\leq N-1}$ for its $M \times N$ leading principal submatrix. When $M = N$, we use X_N instead of $X_{M\times N}$. Then observe that the identity (2.2) for infinite matrices is equivalent to the sequence of identities

$$\widehat{A}_{M\times(N+1)} \;=\; Q_{M\times\infty} \left[\begin{array}{c|c} I_1 & \mathbf{0} \\ \hline \mathbf{0} & \widehat{A}_{\infty\times N} \end{array}\right] \tag{2.11}$$

for all its leading principal submatrices. In particular, when Q is lower-Hessenberg the following identities suffice

$$\widehat{A}_{(N+1)\times(N+2)} \;=\; Q_{(N+1)\times(N+2)} \left[\begin{array}{c|c} I_1 & \mathbf{0} \\ \hline \mathbf{0} & \widehat{A}_{(N+1)} \end{array}\right] , \tag{2.12}$$

and if Q is lower-triangular, we may further simplify to

$$\widehat{A}_{N+1} \;=\; Q_{N+1} \left[\begin{array}{c|c} I_1 & \mathbf{0} \\ \hline \mathbf{0} & \widehat{A}_{N} \end{array}\right] . \tag{2.13}$$

Proof of Theorem 2.3. *(a)* This is immediate from the definition in (2.1).

(b) Here the matrix Q is lower triangular. Define an $N\times N$ matrix $B_{/N} = (b_{ij}^{(N)})_{0\leq i,j\leq N-1}$ by

$$b_{ij}^{(N)} \;=\; \widehat{a}_{N-1-i,j-i} \,. \tag{2.14}$$

That is, the rows of $B_{/N}$ are the rows of A_N taken in reverse order (i.e. interchanging up with down), right-justified in a matrix of N columns. Note in particular that the top row of $B_{/N}$ is the same as the bottom row of A_N, and that

$$B_{/N+1} \;=\; \left[\begin{array}{c|c} \multicolumn{2}{c}{\widehat{\boldsymbol{a}}_N} \\ \hline \mathbf{0} & B_{/N} \end{array}\right] \tag{2.15}$$

where $\widehat{\boldsymbol{a}}_N \stackrel{\text{def}}{=} (\widehat{a}_{Nj})_{0\le j\le N}$ = bottom row of $\widehat{A}_{N+1}$ = top row of $B_{/N+1}$. We then have

$$\left[\begin{array}{c} \widehat{A}_{N+1} \\ \hline B_{/N+1} \end{array}\right] = \left[\begin{array}{c|c} Q_{N+1} & \\ \hline \boldsymbol{q}_N & \\ \hline & I_N \end{array}\right] \left[\begin{array}{c|c} 1 & \\ \hline & \widehat{A}_N \\ \hline & B_{/N} \end{array}\right] \tag{2.16}$$

where the blank entries are zero, and $\boldsymbol{q}_N \stackrel{\text{def}}{=} (q_{N,j})_{0\le j\le N}$ = bottom row of Q_{N+1}. It follows by induction that if Q is TP_r, then so is $\left[\begin{array}{c} \widehat{A}_N \\ \hline B_{/N} \end{array}\right]$ for all $N \ge 0$, and hence so are $B_{/N}$ for all $N \ge 0$.

Now let $R_N = (\delta_{i+j,N-1})_{0\le i,j\le N-1}$ be the $N \times N$ matrix with 1 on the antidiagonal and 0 elsewhere. Left-multiplication (resp. right-multiplication) of an $N \times N$ matrix by R_N reverses the order of its rows (resp. columns). In particular, when $B_{/N}$ is TP_r, then so is $R_N B_{/N} R_N$, which has matrix elements

$$(R_N B_{/N} R_N)_{ij} \;=\; b^{(N)}_{N-1-i,N-1-j} \;=\; \widehat{a}_{i,i-j} \;=\; (\overleftarrow{\widehat{A}})_{ij} \tag{2.17}$$

for $0 \le i, j \le N-1$. Therefore, if Q is TP_r, then so is $(\overleftarrow{\widehat{A}})_N$ for all $N \ge 0$, and hence so is $\overleftarrow{\widehat{A}}$.

(c) Here we assume Q to be lower-Hessenberg. First consider the matrix $\left[\begin{array}{c|c} \multicolumn{2}{c}{Q_{N\times(N+R)}} \\ \hline \mathbf{0} & I_R \end{array}\right]$; the block I_R here starts at column $N+1$ which is the last column until which $Q_{N\times(N+R)}$ has a non-zero entry. If Q is totally positive of order r, so is this matrix.

Now consider the following matrix

$$M_{N,R} \stackrel{\text{def}}{=} \left[\begin{array}{c|c} \multicolumn{2}{c}{Q_{N\times(N+R)}} \\ \hline \mathbf{0} & I_R \end{array}\right] \left[\begin{array}{c|cc} I_1 & & \\ \hline & \multicolumn{2}{c}{Q_{N\times(N+R-1)}} \\ \hline & \mathbf{0} & I_{R-1} \end{array}\right] \left[\begin{array}{c|cc} I_2 & & \\ \hline & \multicolumn{2}{c}{Q_{N\times(N+R-2)}} \\ \hline & \mathbf{0} & I_{R-2} \end{array}\right] \cdots \left[\begin{array}{c|c} I_R & \\ \hline & Q_{N\times(N+1)} \end{array}\right]. \tag{2.18}$$

In this product, each individual matrix has shape $(N+R)\times(N+R)$ except the rightmost one which has shape $(N+R)\times(N+R+1)$.

We claim that the following identity holds:

$$\bigl(T_\infty((\widehat{a}_{N-1,k})_{k\ge 0})^{\mathrm{T}}\bigr)_{R+1} = \; M_{N,R}\begin{bmatrix} N, N+1, \ldots, N+R \\ 0, 1, \ldots, R \end{bmatrix}, \tag{2.19}$$

where the right-hand side denotes the bottom left $(R+1)\times(R+1)$ submatrix of $M_{N,R}$. Note that (2.19) is equivalent to showing that for all i, j satisfying $N \le i \le N+R$, and $0 \le j \le R$, the entry $(M_{N,R})_{i,i-N+j} = \widehat{a}_{N-1,j}$.

We use (2.18) to compute the entry $i, i+j$ of the matrix $M_{N,R}$. Notice that in this computation, we may ignore the first $i-N$ factors on the right-hand side of (2.18) as the lower right portion below the entry i, i is just the identity matrix. This gives us

$$
\begin{align}
(M_{N,R})_{i,i-N+j} &= \left(\left[\begin{array}{c|cc} I_{i-N} & & \\ \hline & \multicolumn{2}{c}{Q_{N\times(2N+R-i)}} \\ \hline & \multicolumn{1}{|c|}{\mathbf{0}} & I_{R+N-i} \end{array}\right] \cdots \left[\begin{array}{c|c} I_R & \\ \hline & Q_{N\times(N+1)} \end{array}\right] \right)_{i,i-N+j} \tag{2.20a} \\
&= \left(\left[\begin{array}{c|c} \multicolumn{2}{c}{Q_{N\times(2N+R-i)}} \\ \hline \mathbf{0} & I_{R+N-i} \end{array}\right] \cdots \left[\begin{array}{c|c} I_{R+N-i} & \\ \hline & Q_{N\times(N+1)} \end{array}\right] \right)_{N,j} \tag{2.20b} \\
&= \widehat{a}_{N-1,j} \; . \tag{2.20c}
\end{align}
$$

where the second equality is obtained by taking the lower right $(R+2N-i)\times(R+2N-i)$ submatrix of each individual factor except for the last one where we take the lower-right $(R+2N-i)\times(R+2N-i+1)$. The third equality is simply (2.1). This establishes the claimed identity (2.19).

Finally, if Q is TP_r, so is $M_{N,R}$ for every N, R and so must be the Toeplitz matrix $T_\infty((\widehat{a}_{N-1,k})_{k\ge 0})$. $\square$

Remark. The working of Equation (2.20) is inspired by the combinatorial arguments used in the proof of [5, Theorem 1.4]. Notice that the their theorem requires the left production matrix to be lower triangular. We have lifted this requirement on all of the statements except for (b). $\blacksquare$

PLEASE CHECK IF I HAVE DONE THE PROOF CORRECTLY!!!!!!!!

2.3 Relation between left and right production matrices

The method of production matrices [9,10] has become in recent years an important tool in enumerative combinatorics. This method has also been successfully employed to obtain several total positivity results, see for example, **CITE!!!!!!!**. In this section we briefly recall production matrices following [20, sections 2.2 and 2.3] and then show their relation to our left-production matrices. To avoid confusion with left-production matrices, we will refer to them as right-production matrices in this paper.

Let $P = (p_{ij})_{i,j\ge 0}$ be an infinite matrix with entries in a commutative ring R, and assume that P is either row-finite or column-finite (so that powers of P are well-defined). Now define a matrix $A = (a_{nk})_{n,k\ge 0}$ by

$$
a_{nk} \;=\; (P^n)_{0k} \tag{2.21}
$$

(note in particular that $a_{0k} = \delta_{k0}$). We call P the ***right-production matrix*** and A the ***right-output matrix***, and we write $A = \mathcal{O}(P)$. Another equivalent formulation of (2.21), is the following recurrence

$$a_{nk} \;=\; \sum_{i=0}^{\infty} a_{n-1,i}\, p_{ik} \;, \tag{2.22}$$

with the initial condition $a_{0k} = \delta_{k0}$. This is equivalent to $\Delta A = AP$, where recall that Δ is the matrix with 1 on the superdiagonal and 0 elsewhere. Note that if P is lower-Hessenberg (i.e. vanishes above the first superdiagonal), then $\mathcal{O}(P)$ is lower-triangular; this is the most common case, and will be the case here. Conversely, when A is lower-triangular with invertible diagonal entries and $a_{00} = 1$, then there exists a unique production matrix generating A, and it is obtained as

$$P \;=\; A^{-1}\Delta A\,. \tag{2.23}$$

Let us define the ***augmented right-production matrix*** to be the

$$\widetilde{P} \;\stackrel{\text{def}}{=}\; \left[\begin{array}{c|c} 1 & \mathbf{0} \\ \hline \multicolumn{2}{c}{P} \end{array}\right]. \tag{2.24}$$

The relation $\Delta A = AP$ can be rewritten in terms of $\widetilde{P}$ as

$$A \;=\; \left[\begin{array}{c|c} 1 & \mathbf{0} \\ \hline \mathbf{0} & A \end{array}\right] \widetilde{P} \tag{2.25}$$

Iterating this, we can express the right-output matrix A in a form that is similar to equation (2.1):

$$A \;=\; \cdots \left[\begin{array}{c|c} I_3 & \mathbf{0} \\ \hline \mathbf{0} & \widetilde{P} \end{array}\right] \left[\begin{array}{c|c} I_2 & \mathbf{0} \\ \hline \mathbf{0} & \widetilde{P} \end{array}\right] \left[\begin{array}{c|c} I_1 & \mathbf{0} \\ \hline \mathbf{0} & \widetilde{P} \end{array}\right] \widetilde{P}\;. \tag{2.26}$$

Equation (2.26) justifies why we call P as the right-production matrix of A.

Finally, we have the following relation between the left and right production matrices.

Proposition 2.4. *Let P be a right-production matrix and let A be its right-output matrix. Then the matrix $\widetilde{P}^{\mathrm{T}}$ is a left-production matrix and its left-output matrix is A^{T}.*

Proof. We tranpose both sides of equation (2.26). $\square$

Thus, the left-production matrix of a matrix A is the transpose of the right-production matrix of its transpose A^{T}.

We finally end this section by stating an extremely shocking corollary.

Corollary 2.5. *Let P be an upper triangular matrix and $A = \mathcal{O}(P)$ be its right output matrix. If P is totally positive of order r, then each column sequence of A is Toeplitz totally positive of order r.*

Proof. Consider the augmented right-production matrix $\widetilde{P}$ given by (2.25); its transpose $\widetilde{P}^{\mathrm{T}}$ is lower Hessenberg. The left output matrix of $\widetilde{P}^{\mathrm{T}}$ is A^{T} by Proposition 2.4. The proof now follows from Theorem 2.3(c). $\square$

We now state why this corollary is shocking. By virtue of [17, Theorem 9.7], we know that if a right production matrix (that is row-finite or column-finite) is totally positive, then its zeroth column is Hankel totally positive. However, we just noticed in Corollary 2.5 that for an upper triangular right production matrix, not only its zeroth column, but all columns are Toeplitz totally positive. However, on further inspection we can notice that for such a right production matrix, the column 0 of its right output matrix is simply $(p_{00}^n)_{n\geq 0}$ which is indeed both Hankel and Toeplitz totally positive.

3 Generalizations of some results of Bränden–Saud Maia Leite using the theory of left and right production matrices

In this section, we will study some notions developed by Bränden and Saud Maia Leite recently in [1] using our theory of left and right production matrices. We will work over an arbitrary partially ordered commutative ring and will thus be able to provide vast generalizations of their results [1, Theorems 3.7 and 4.3]. In Section 3.1 we study some sequences generated by a totally positive lower triangular matrix which turn out to be Toeplitz totally positive; in particular we generalize [1, Theorem 4.3]. Then in Section 3.2 we introduce h-polynomials and recall *chain polynomials* introduced by Bränden and Saud Maia Leite and then generalize their [1, Theorem 3.7].

3.1 Some Toeplitz totally positive sequences generated from a totally positive lower triangular matrix

Let R be a lower-triangular matrix with entries in a partially ordered commutative ring. For any $i, j \in \mathbb{N}$ satisfying $i \geq j$, we consider the sequence $\big((R^k)_{ij}\big)_{k\geq 0}$. We first recall [1, Theorem 4.3(b)]:

Theorem 3.1 ([1, Theorem 4.3(b)]). *Let R be a unit lower-triangular matrix of real numbers. If R is totally positive then for each given i, j satisfying $i \geq j$, the sequence $\big((R^k)_{ij}\big)_{k\geq 0}$ is Toeplitz totally positive.*

This result was proved using the interlacing of zeroes of chain polynomials (we will see these in the next subsection). We will now generalize their result: we will lift the restriction that the diagonal elements need to be 1 and our result will hold over any partially ordered commutative ring and for total positivity of any order. Our proof will be algebraic.

Theorem 3.2. *Let $R = (r_{n,k})_{n,k\geq 0}$ be a lower triangular matrix with elements in a partially ordered commutative ring R, and let $i, j \in \mathbb{N}$ be fixed satisfying $i \geq j$. Then* *if R is totally positive of order r, the sequence $\big((R^k)_{ij}\big)_{k\ge 0}$ is Toeplitz totally positive of order r.*

Before proving this theorem, we first define the matrix $B := (b_{nk})_{n,k\ge 0}$ by

$$b_{nk} \;=\; (R^k)_{n,0}\,. \tag{3.1}$$

We then establish the following lemma.

Lemma 3.3. *Given a row-finite or column-finite matrix $R = (r_{nk})_{n,k\ge 0}$, we set $\widehat{R}$ to be the following matrix $\widehat{R} := (\mathbf{e}_{00} + R\Delta)$, i.e.,*

$$\widehat{R} \;=\; \left[\begin{array}{c|c} 1 & R \\ \hline & \\ \mathbf{0} & \end{array}\right]. \tag{3.2}$$

Then $\widehat{R}$ when considered as a left-production matrix gives $B = \big((R^k)_{n,0}\big)_{n,k\ge 0}$ as its left output matrix.

Proof. We show that the entries $b_{n,k}$ satisfy the recurrence (2.3). First notice that by definition $b_{n0} = \delta_{n0} = \Big(\widehat{R}\Big)_{n0}$ which is the initial condition that we want. Finally, we notice that

$$b_{n,k+1} \;=\; (R\cdot R^k)_{n,0} \;=\; \sum_i r_{ni}\, b_{i,k} \;=\; \sum_{i=0}^{\infty} \Big(\widehat{R}\Big)_{n,i+1}\, b_{i,k} \tag{3.3}$$

which is what we want. $\square$

We are now ready to prove Theorem 3.2.

Proof of Theorem 3.2. If R is totally positive of order r, so is $\widehat{R}$. When $j > 0$, we follow Brändén and Saud Maia Leite and recall their observation $(R^k)_{ij} = (S^k)_{i-j,0}$ where S is the matrix obtained from R by deleting the j initial rows and columns of R; see [1, proof of Theorem 4.3]. Thus, it suffices to only deal with $j = 0$. Here, the desired result follows by using Lemma 3.3 together with Theorem 2.3(c) when applied to row i of the matrix B. $\square$

3.2 h-polynomials and chain polynomials

3.2.1 h-polynomials

Let $Q = (q_{nk})_{n,k\ge 0}$ be a lower-Hessenberg left production matrix and let $\widehat{A} = (\widehat{a}_{nk})_{n,k\ge 0}$ be its corresponding left output matrix. We recall that if the super-diagonal entries of $(q_{n,n+1})_{n\ge 0}$ are non-zero, then $\widehat{A}$ will be a dense matrix. Let $\widehat{A}_n(x) := \sum_{k=0}^{\infty} \widehat{a}_{nk}\, x^k$ denote the n-th row-generating o.g.f. of the matrix $\widehat{A}$.

We now rewrite the recurrence (2.5) in terms of $\widehat{A}_n(x)$ to obtain

$$\widehat{A}_0(x) \;=\; \frac{q_{00}}{1 \,-\, q_{01}x} \tag{3.4a}$$

$$\widehat{A}_{n+1}(x) \;=\; q_{n+1,0} \,+\, x \sum_{i=0}^{n+1} q_{n+1,i+1}\, \widehat{A}_i(x) \qquad \text{for } n \geq 0\ . \tag{3.4b}$$

From this rewriting we immediately get the following proposition.

Proposition 3.4. *Let $Q = (q_{nk})_{n,k\geq 0}$ be a lower-Hessenberg left production matrix and let $\widehat{A} = (\widehat{a}_{nk})_{n,k\geq 0}$ be its corresponding left output matrix. There exists a unique sequence of polynomials $(h_n(x))_{n\geq 0}$ such that the degree of the polynomial $h_n(x)$ is at most n, and the n-th row-generating o.g.f. of the matrix $\widehat{A}$ satisfies*

$$\widehat{A}_n(x) \;=\; \frac{h_n(x)}{\prod_{i=0}^{n}(1 \,-\, xq_{i,i+1})}\ . \tag{3.5}$$

The proof of Proposition 3.4 is straightforward from the recurrence (3.4).

Definition 3.5. *For a given lower-Hessenberg left production matrix Q, its corresponding* ***sequence of h-polynomimals*** *is the sequence $(h_n(x))_{n\geq 0}$ given by Proposition 3.4.*

An immediate corollary of the existence of h-polynomials is the following generalization of [1, Theorem 4.3(a)].

Corollary 3.6. *Let the underlying ring R be a commutative ring with unity that is a $\mathbb{Q}$-algebra. Let $Q = (q_{n,k})_{n,k\geq 0}$ be a unit lower-Hessenberg matrix and let $\widehat{A} = (\widehat{a}_{nk})_{n,k\geq 0}$ be its corresponding left output matrix. We then have:*

- *(a) For each $n \in \mathbb{N}$, there exists a polynomial $P_n : \mathbb{N} \to R$ of degree n such that entries of the n-th row of $\widehat{A}$ are obtained by evaluating this polynomial, i.e. $P_n(k) = \widehat{a}_{n,k}$.*
- *(b) Let $i, j \in \mathbb{N}$ be fixed satisfying $i \geq j$, and let $R = (r_{n,k})_{n,k\geq 0}$ be a unit lower-triangular matrix. The sequence $\big((R^k)_{i,j}\big)_{k\geq 0}$ is obtained by evaluating a polynomial of degree $i - j$.*

Proof. *(a)* From the existence of h-polynomials in Proposition 3.4 and the fact that the super diagonal entries are $q_{i,i+1} = 1$ for all $i \geq 0$, we have $\widehat{A}_n(x) \;=\; \dfrac{h_n(x)}{(1 \,-\, x)^{n+1}}$. Thus, from a well-known result **FIND CORRECT REFERENCE FOR THE MOST GENERAL RING!!!!!**, we get that the n-th row sequence is obtained by evaluating a polynomial $P_n : \mathbb{N} \to R$ of degree at most n. Finally, we will soon see in Lemma 3.7/Proposition 3.8 that $h_n(1) \neq 0$ which shows that P_n has degree exactly n.

(b) This follows by combining (a) with Lemma 3.3. $\square$

Remark. If a polynomial f of degree d satisfies $\sum_{n=0}^{\infty} f(n)x^n = \dfrac{g(x)}{(1-x)^{d+1}}$ for some polynomial g of degree d, then the polynomial $g(x)$ is frequently called the f-Eulerian polynomial, see [14] and references therein. ■

Our first observation is that the coefficients of h-polynomials can be expressed in terms of minors of the matrix left production matrix Q. Here, for a finite set $S \subset \mathbb{N}$, we use the notation $S + a := \{s + a | s \in S\}$.

Lemma 3.7. *Let $Q = (q_{n,k})_{n,k\geq 0}$ be a unit lower-Hessenberg left production matrix and let $(h_n(x))_{n\geq 0}$ denote its sequence of h-polynomials. The coefficients of the polynomial $h_n(x)$ can be expressed as*

$$[x^k]\, h_n(x) \;=\; \sum_{S\in\binom{[0,n-1]}{k}} Q\left[\begin{array}{c} S\cup\{n\} \\ \{0\}\cup(S+1)\end{array}\right] \tag{3.6}$$

where the sum is over all subsets S of size k of the set $\{0,\ldots,n-1\}$ and the summand is the minor of the matrix Q corresponding to rows $S\cup\{n\}$ and columns $\{0\}\cup(S+1)$.

Proof. We prove this using induction. For $n = 0$, $h_0(x) = q_{00}$ which is exactly the minor of Q corresponding to $S = \varnothing, n = 0$. Let us now assume that the formula holds for $n \geq 0$ and we will establish it for $n + 1$.

We rewrite (3.4b) as $\widehat{A}_{n+1}(x)(1 - q_{n+1,n+2}x) = q_{n+1,0} + x \sum_{i=0}^{n} q_{n+1,i+1}\, \widehat{A}_i(x)$ and multiply $\prod_{i=0}^{n}(1 - q_{i,i+1}x)$ on both sides to obtain

$$\begin{aligned}
h_{n+1}(x) &= q_{n+1,0}\prod_{i=0}^{n}(1-q_{i,i+1}x) \;+\; x\sum_{i=0}^{n} q_{n+1,i+1}\,h_i(x)\prod_{j=i+1}^{n}(1-q_{i,i+1}x)\\
&= (1-q_{n,n+1}x)\left(q_{n+1,0}\prod_{i=0}^{n-1}(1-q_{i,i+1}x) \;+\; x\sum_{i=0}^{n-1} q_{n+1,i+1}\,h_i(x)\prod_{j=i+1}^{n}(1-q_{i,i+1}x)\right)\\
&\quad + xq_{n+1,n+1}h_n(x)\\
&= h_n(x)\big|_{q_{n,i}\mapsto q_{n+1,i}}\,(1-q_{n,n+1}x) \;+\; xq_{n+1,n+1}h_n(x)\;.
\end{aligned} \tag{3.7}$$

Now we extract the coefficient of x^k on both sides of this equation to obtain

$$\begin{aligned}
[x^k]\,h_{n+1}(x) &= \sum_{S\in\binom{[0,n-1]}{k}} Q\left[\begin{array}{c} S\cup\{n+1\} \\ \{0\}\cup(S+1)\end{array}\right] - \sum_{S\in\binom{[0,n-1]}{k-1}} q_{n,n+1}Q\left[\begin{array}{c} S\cup\{n+1\} \\ \{0\}\cup(S+1)\end{array}\right]\\
&\quad + \sum_{S\in\binom{[0,n-1]}{k-1}} q_{n+1,n+1}Q\left[\begin{array}{c} S\cup\{n\} \\ \{0\}\cup(S+1)\end{array}\right]\\
&= \sum_{S\in\binom{[0,n-1]}{k}} Q\left[\begin{array}{c} S\cup\{n+1\} \\ \{0\}\cup(S+1)\end{array}\right] + \sum_{S\in\binom{[0,n-1]}{k-1}} Q\left[\begin{array}{c} S\cup\{n,n+1\} \\ \{0,n+1\}\cup(S+1)\end{array}\right]\\
&= \sum_{S\in\binom{[0,n]}{k}} Q\left[\begin{array}{c} S\cup\{n,n+1\} \\ \{0,n+1\}\cup(S+1)\end{array}\right]
\end{aligned} \tag{3.8}$$

This finishes the proof. □

An immediate consequence of Lemma 3.7 is the following observation.

Proposition 3.8. *Let $Q = (q_{n,k})_{n,k\geq 0}$ be a unit lower-Hessenberg left production matrix and let $(h_n(x))_{n\geq 0}$ denote its sequence of h-polynomials. If Q is totally positive, then for all $n \in \mathbb{N}$, the coefficients of $h_n(x)$ are positive in the underlying partially ordered commutative ring.*

Let the left production matrix $Q = \widehat{R}$ as in the setting of Lemma 3.3. Furthermore, assume that R is a unit-lower triangular matrix, making Q unit-lower Hessenberg, and also that the entries are over the reals. In this case, Bränden and Saud Maia Leite showed that if the matrix R is totally positive then the h-polynomials are negative-real-rooted [1, Proof of Theorem 4.3]. In particular, this implies that the coefficients of h_n are non-negative. Motivated by their result we leave the following open problem:

Problem 3.9. *Let $Q = (q_{nk})_{n,k\geq 0}$ be a lower-Hessenberg left production matrix and let $(h_n(x))_{n\geq 0}$ be its corresponding sequence of h-polynomials. Then find necessary and sufficient conditions on the matrix Q such that for each $n \geq 0$, the coefficient sequence $\big([x^k]h_n(x)\big)_{k\geq 0}$ is Toeplitz totally positive.*

At this moment, we are unable to address this question.

3.2.2 Chain polynomials

For a given unit-lower triangular matrix R, Bränden and Saud Maia Leite [1] recently introduced the notion of *chain polynomials* which we now recall.

Definition 3.10. *For a unit-lower triangular matrix $R = (r_{n,k})_{n,k\geq 0}$, the chain polynomials associated to R is the sequence $(p_n(x))_{n\geq 0}$ defined by $p_0(x) = 1$, and for $n \geq 0$,*

$$p_{n+1}(x) \stackrel{\text{def}}{=} x \sum_{i=0}^{n} r_{n+1,i}\, p_i(x) \; . \tag{3.9}$$

The chain coefficient matrix C is the matrix whose (n,k)-th entry is the coefficient $[x^k]p_n(x)$.

Our first observation here is that this construction of Bränden and Leite can be rewritten in our theory of left-production matrices as follows:

Proposition 3.11 (Chain polynomials from left output matrix)**.** *Let $\widetilde{R}$ denote the matrix $\widetilde{R} = (\mathbf{e}_{00} + (R - I)\,\Delta)$. (Here $\widetilde{R}$ is lower triangular, and (n,k)-th entry for $n \geq 1$ and $1 \leq k \leq n$ is $r_{n,k-1}$.) Considering the matrix $\widetilde{R}$ as a left production matrix gives us the chain coefficient matrix C as its left output matrix.*

Proof. Let $\widetilde{A}$ denote the left output matrix of $\widetilde{R}$ and let $\widetilde{A}_n(x)$ denote the n-th row-generating polynomial of $\widetilde{A}$. We rewrite the recurrence (3.4) in this setting to obtain

$$\widetilde{A}_0(x) = 1 \tag{3.10a}$$

$$\widetilde{A}_{n+1}(x) = x \sum_{i=0}^{n} r_{n+1,i} \widetilde{A}_i(x) \; , \tag{3.10b}$$

which is the same recurrence as in 3.9 and thus, we conclude $C = \widetilde{A}$. $\square$

Before proceeding we recall one of the main results in [1].

Theorem 3.12 ([1, Theorem 3.7]). *Let $R = (r_{n,k})_{n,k\geq 0}$ be a unit lower-triangular matrix with real entries. If R is totally positive, then for each $n \in \mathbb{N}$, the zeros of the chain polynomial $p_n(x)$ are real and located in the interval $[-1, 0]$.*

Their theorem is extremely surprising from our point of view! Notice that according to Theorem 2.3(c) and Proposition 3.11, if the matrix $\widetilde{R}$ is totally positive, which is equivalent to the total positivity of $R - I$, it would imply that the chain polynomials $p_n(x)$ are real-rooted with zeros in the interval $(-\infty, 0]$. However, here the hypothesis is that R, and not $R - I$, is totally positive. Indeed, for a unit lower triangular matrix of real numbers A, the total positivities of the matrix A and $A - I$ are not related to each other. For example, let A be the following 5×5 matrix

$$A = \begin{bmatrix} 1 & & & & \\ 1 & 1 & & & \\ 1 & 2 & 1 & & \\ 1 & 3 & 3 & 1 & \\ 0 & 1 & 2 & 1 & 1 \end{bmatrix}. \tag{3.11}$$

Even though the matrix A is totally positive, the matrix $A - I$ is not. In fact, the only real number a such that $A - aI$ is totally positive is when $a = 0$.

NOW WHAT???????? It seems the following relation between h-polynomials and chain polynomials holds for R unit lower-triangular: $p_n(x) = (1+x)^n h_n(x/(1+x))$. Thus, p_n is always TP_1 whenever R is TP.

3.3 Toeplitz left production matrices

We first establish the following lemma.

Lemma 3.13. *Consider the sequence $(f_n)_{n\geq 0}$ with the ordinary generating function $f(t) = \sum_{n=0}^{\infty} f_n t^n$, and also let $F = (f_{n-k})_{n,k\leq 0}$ be its lower triangular Toeplitz matrix. Consider the following formal power series*

$$\frac{f(t)}{1 - xtf(t)} = \sum_{n=0}^{\infty} \widehat{a}_n(x) t^n\,, \tag{3.12a}$$

$$\frac{1}{1 - xtf(t)} = \sum_{n=0}^{\infty} \widehat{b}_n(x) t^n\,. \tag{3.12b}$$

Here the coefficients $\widehat{a}_n(x), \widehat{b}_n(x)$ of the two series on the right hand side are polynomials in the variable x, respectively. Then we have

(a) *The polynomials $\widehat{a}_n(x)$ are the row-generating polynomials of the left output matrix of F considered as a left production matrix.*

(b) Let $\widehat{F}$ be the matrix

$$\widehat{F} = \left[\begin{array}{c|c} 1 & \\ \hline & F \end{array}\right] . \tag{3.13}$$

The polynomials $\widehat{b}_n(x)$ are the row-generating polynomials of the left output matrix of $\widehat{F}$ considered as a left production matrix.

PROOF. Let A and $\widehat{A}$ be the left-output matrices of F and $\widehat{F}$, respectively. As $F, \widehat{F}$ are lower triangular so are $A, \widehat{A}$, respectively. Also, let $A_n(x), \widehat{A}_n(x)$ denote the row-generating polynomial of the n-th row of $A, \widehat{A}$, respectively. We rewrite the recurrence (3.4) in this setting to obtain for F

$$A_0(x) = f_0 \tag{3.14a}$$

$$A_{n+1}(x) = f_{n+1} + x \sum_{i=0}^{n} f_{n-i} A_i(x) , \tag{3.14b}$$

and for $\widehat{F}$, we obtain

$$\widehat{A}_0(x) = 1 \tag{3.15a}$$

$$\widehat{A}_{n+1}(x) = x \sum_{i=0}^{n} f_{n-i} \widehat{A}_i(x) . \tag{3.15b}$$

Rewriting (3.16a) as $f(t) = (1 - xtf(t))(\sum_{n=0}^{\infty} \widehat{a}_n(x)t^n)$, and (3.16b) as $1 = (1 - xtf(t))(\sum_{n=0}^{\infty} \widehat{b}_n(x)t^n)$, and comparing the coefficients of t^0 and t^{n+1} from both sides in each equation establishes $\widehat{a}_n(x) = A_n(x)$ and $\widehat{b}_n(x) = \widehat{A}_n(x)$ as they satisfy the same recurrences. □

As a consequence of Lemma 3.13, we have the following result that is motivated by [1, Theorem 4.4].

Theorem 3.14. *Consider the sequence $(f_n)_{n\geq 0}$ with the ordinary generating function $f(t) = \sum_{n=0}^{\infty} f_n t^n$, and also let $F = (f_{n-k})_{n,k\leq 0}$ be its lower triangular Toeplitz matrix. Consider the following formal power series*

$$\frac{f(t)}{1 - xtf(t)} = \sum_{n=0}^{\infty} \widehat{a}_n(x) t^n , \tag{3.16a}$$

$$\frac{1}{1 - xtf(t)} = \sum_{n=0}^{\infty} \widehat{b}_n(x) t^n . \tag{3.16b}$$

If $(f_n)_{n\geq 0}$ is Toeplitz totally positive of order r, then we have

(a) For each n the coefficient sequence $\big([x^i]\, \widehat{a}_n(x)\big)_{i\geq 0}$ is Toeplitz totally positive of order r.

(b) For each n the coefficient sequence $\big([x^i]\, \widehat{b}_n(x)\big)_{i\geq 0}$ is Toeplitz totally positive of order r.

PROOF. The proof follows by combining Lemma 3.13 and Theorem 2.3(c). □

4 Application: total positivity of the matrices $\boldsymbol{T}(0, c, d, e)$ and $\boldsymbol{T}(a, c, 0, e)$ in [4]

In [4], we considered a lower triangular matrix $\boldsymbol{T} = \big(T(n,k)\big)_{n,k\geq 0}$ whose entries are defined by the following linear recurrence,

$$T(n,k) \;=\; [a(n-k)+c]\,T(n-1,k-1) + (dk+e)\,T(n-1,k) + [f(n-2)+g]\,T(n-2,k-1) \tag{4.1}$$

for $n \geq 1$, with initial conditions $T(0,k) = \delta_{k0}$ and $T(-1,k) = 0$. We conjectured [4, Conjecture 1.4] that this matrix is totally positive in all six indeterminates a, c, d, e, f, g. This conjecture generalises Brenti's Eulerian triangle conjecture [3] which can be seen by substituting $a = c = d = e = 1$ and $f = g = 0$ in the matrix $\boldsymbol{T}$.

Our main result in [4, Theorems 1.5/3.2] was proving the total positivity of the matrix $\boldsymbol{T}(a, c, 0, e)$ which is obtained by substituting $d = f = g = 0$ in the matrix $\boldsymbol{T}$; we proved this by constructing an explicit *planar network* and we showed that its *path matrix* is the the matrix $\boldsymbol{T}(a, c, 0, e)$. We will now apply the method of left-production matrices and give a new proof of this result. In [5, sec. 4], The left-production-matrix method was used to prove the total positivity of the matrix $\boldsymbol{T}(1, 1, 0, 1)$, (the reversed Stirling subset matrix).

We first consider the matrix $\boldsymbol{T}(0, c, d, e)$; this the matrix $\boldsymbol{T}$ along with the substitution $a = f = g = 0$. In this special case, the recurrence (4.1) becomes purely k-dependent and is therefore coefficientwise totally positive [2]. Also, notice that $\overleftarrow{\boldsymbol{T}(0, c, d, e)} = \boldsymbol{T}(d, e, c, 0)$. We will now construct the left-production matrix of the matrix $\boldsymbol{T}(0, c, d, e)$ and prove that it is totally positive; we then use Theorem 2.3 to infer the total positivity of both matrices $\boldsymbol{T}(0, c, d, e)$ and $\boldsymbol{T}(a, c, e, 0)$ (after a relabeling of parameters).

We first introduce some notation. Let T_ξ be the ***lower-triangular Toeplitz matrix of powers***, with matrix elements

$$(T_\xi)_{nk} \;=\; \begin{cases} \xi^{n-k} & \text{if } n \geq k \\ 0 & \text{otherwise} \end{cases} . \tag{4.2}$$

Next, we consider the ***weighted binomial matrix*** $B_{x,y} = \big((B_{x,y})_{nk}\big)_{n,k\geq 0}$ defined by

$$(B_{x,y})_{nk} \;=\; \binom{n}{k}\, x^{n-k}\, y^k\ . \tag{4.3}$$

We also write B_x as a shorthand for $B_{x,1}$. It is well known that both matrices T_ξ and $B_{x,y}$ are coefficientwise totally positive in the variables ξ and x, y, respectively.

We are now ready to state the left-production matrix of $\boldsymbol{T}(0, c, d, e)$.

Proposition 4.1 (Left-production matrix for $\boldsymbol{T}(0, c, d, e)$)**.** *The matrix*

$$Q \;\stackrel{\text{def}}{=}\; T_e \left[\begin{array}{c|c} 1 & \mathbf{0} \\ \hline \mathbf{0} & c\,B_d \end{array}\right] \tag{4.4}$$

satisfies

$$\boldsymbol{T}(0,c,d,e) \;=\; Q \left[\begin{array}{c|c} 1 & \mathbf{0} \\ \hline \mathbf{0} & \boldsymbol{T}(0,c,d,e) \end{array}\right] . \tag{4.5}$$

That is, the left-production matrix Q defined in (4.4) has $\boldsymbol{T}(0,c,d,e)$ as its left-output matrix.

We will provide two proofs of Proposition 4.1 in Section 4.1.

From the total positivity of the matrices T_e and B_d, we can immediately conclude:

Proposition 4.2. *The matrix Q defined in (4.4) is coefficientwise totally positive in the indeterminates c,d,e.*

Finally, we use Theorem 2.3 to obtain the following theorem:

Theorem 4.3. *The matrices $\boldsymbol{T}(0,c,d,e)$ and $\boldsymbol{T}(d,e,c,0)$ are totally positive in the indeterminates a,c,d,e. Furthermore, for each row $n \geq 1$, the infinite upper triangular Toeplitz matrix $T_\infty((T(n,k))_{k\geq 0}) = (T(n,i-j))_{i,j\geq 0}$ is coefficientwise Toeplitz-totally positive in the indeterminates a,c,e.*

Remark. The coefficientwise Toeplitz-total positivity of the sequence $T_\infty((T(n,k))_{k\geq 0})$ is a vast generalization of the real-rootedness of the Stirling subset polynomials. ■

4.1 Proofs of Proposition 4.1

In this section, we will now provide two proofs of Proposition 4.1, one algebraic and one bijective.

We require Lemmas 4.4–4.6 for our first proof.

Lemma 4.4 (Factorization of $B_{x,y}$ in terms of T_x)**.** *We have*

$$B_{x,y} \;=\; T_x \left[\begin{array}{c|c} 1 & \mathbf{0} \\ \hline \mathbf{0} & y\,B_{x,y} \end{array}\right] \tag{4.6}$$

and hence

$$B_{x,y} \;=\; T_x \left[\begin{array}{c|c} I_1 & \mathbf{0} \\ \hline \mathbf{0} & y\,T_x \end{array}\right] \left[\begin{array}{c|c} I_2 & \mathbf{0} \\ \hline \mathbf{0} & y\,T_x \end{array}\right] \left[\begin{array}{c|c} I_3 & \mathbf{0} \\ \hline \mathbf{0} & y\,T_x \end{array}\right] \cdots \tag{4.7}$$

where I_k is the $k \times k$ identity matrix.

Proof. Let L_{-x} be the matrix with 1 on the diagonal, $-x$ on the subdiagonal, and 0 elsewhere; we have $T_x = (L_{-x})^{-1}$. Then

$$(L_{-x}B_{x,y})_{nk} \;=\; \binom{n}{k} x^{n-k}\, y^k \;-\; x \binom{n-1}{k} x^{n-1-k}\, y^k \tag{4.8a}$$

$$\;=\; \binom{n-1}{k-1} x^{n-k}\, y^k \;, \tag{4.8b}$$

which is precisely the (n,k) matrix element of $\left[\begin{array}{c|c} 1 & \mathbf{0} \\ \hline \mathbf{0} & y\,B_{x,y} \end{array}\right]$. This proves (4.6). Then (4.7) follows by iteration (note that each matrix element stabilizes after a finite number of factors). □

Lemma 4.5 (Binomial transform)**.** *We have*

$$B_{\xi,\eta}\,\boldsymbol{T}(0,c,d,e) \;=\; \boldsymbol{T}(0,\,\eta c,\,\eta d,\,\eta e+\xi)\;. \tag{4.9}$$

In particular, when $\xi=-e$ and $\eta=1$, we have

$$\boldsymbol{T}(0,c,d,e) \;=\; B_e\,\boldsymbol{T}(0,c,d,0)\;. \tag{4.10}$$

We remark that Lemma 4.5 is (when normalized to $\eta=1$) a special case of [18, Corollary A.11]. For completeness we include the proof:

Proof of Lemma 4.5. Writing $B=B_{\xi,\eta}$ for conciseness, the matrix $\boldsymbol{M}=B_{\xi,\eta}\,\boldsymbol{T}(0,c,d,e)$ satisfies

$$\begin{aligned}
M(n,k) &= \sum_{j=0}^{n} B(n,j)\,T(j,k) && \text{(4.11a)}\\
&= \sum_{j=0}^{n} \big[\xi\,B(n-1,j)+\eta B(n-1,j-1)\big]\,T(j,k) && \text{(4.11b)}\\
&= \xi\,M(n-1,k)+\eta\sum_{j=1}^{n} B(n-1,j-1)\times\Big[c\,T(j-1,k-1) \\
&\qquad\qquad + (dk+e)\,T(j-1,k)\Big] && \text{(4.11c)}\\
&= c\eta\,M(n-1,k-1) \;+\; \big[(dk+e)\eta+\xi\big]\,M(n-1,k)\;, && \text{(4.11d)}
\end{aligned}$$

which implies (4.10). □

Lemma 4.6 (Pulling out a Toeplitz matrix of powers)**.** *We have*

$$\boldsymbol{T}(0,c,d,e) \;=\; T_e\left[\begin{array}{c|c} 1 & \mathbf{0} \\ \hline \mathbf{0} & c\,\boldsymbol{T}(0,c,d,d+e) \end{array}\right]. \tag{4.12}$$

The proof of Lemma 4.6 will be based on Lemma 4.5 combined with the identity (4.7) that relates the e-binomial matrix to the e-Toeplitz matrix of powers.

Proof of Lemma 4.6. As special cases of (4.9), we have

$$B_e\,\boldsymbol{T}(0,c,d,0) \;=\; \boldsymbol{T}(0,c,d,e) \tag{4.13}$$

$$B_e\,\boldsymbol{T}(0,c,d,d) \;=\; \boldsymbol{T}(0,c,d,d+e) \tag{4.14}$$

We further claim that the following identity is also true:

$$\boldsymbol{T}(0,c,d,0) \;=\; \left[\begin{array}{c|c} 1 & \mathbf{0} \\ \hline \mathbf{0} & c\,\boldsymbol{T}(0,c,d,d) \end{array}\right] . \tag{4.15}$$

It is clear that the matrix $\boldsymbol{T}(0,c,d,0)$ can be written as $\boldsymbol{T}(0,c,d,0) = \left[\begin{array}{c|c} 1 & \mathbf{0} \\ \hline \mathbf{0} & \boldsymbol{U} \end{array}\right]$ where $U(n,k) = T(n+1,k+1;\, 0,c,d,0)$. Inserting the recurrence (4.1) and using $U(-1,-1) = T(0,0) = 1$, we obtain

$$U(n,k) \;=\; c\,U(n-1,k-1) \;+\; (dk+d)\,U(n-1,k) \;+\; c\delta_{n0}\delta_{k0} \,. \tag{4.16}$$

This shows that $U(n,k) = c\,T(n,k;\, 0,c,d,d)$ which implies our claim.

Finally, obtain the desired identity as follows

$$\begin{aligned}
&\boldsymbol{T}(0,c,d,e) \\
&\quad= B_e\,\boldsymbol{T}(0,c,d,0) \qquad \text{by (4.13)} && \text{(4.17a)} \\
&\quad= T_e \left[\begin{array}{c|c} 1 & \mathbf{0} \\ \hline \mathbf{0} & B_e \end{array}\right] \left[\begin{array}{c|c} 1 & \mathbf{0} \\ \hline \mathbf{0} & c\,\boldsymbol{T}(0,c,d,d) \end{array}\right] \quad \text{by (4.7) and (4.15)} && \text{(4.17b)} \\
&\quad= T_e \left[\begin{array}{c|c} 1 & \mathbf{0} \\ \hline \mathbf{0} & c\,\boldsymbol{T}(0,c,d,d+e) \end{array}\right] \qquad \text{by (4.14)} \,. && \text{(4.17c)}
\end{aligned}$$

□

We are now ready to prove Proposition 4.1.

Algebraic Proof of Proposition 4.1. Using the identities in Lemmas 4.6 and 4.5, we have

$$\begin{aligned}
\boldsymbol{T}(0,c,d,e) \;&=\; T_e \left[\begin{array}{c|c} 1 & \mathbf{0} \\ \hline \mathbf{0} & c\,\boldsymbol{T}(0,c,d,d+e) \end{array}\right] && \text{(4.18a)} \\
&=\; T_e \left[\begin{array}{c|c} 1 & \mathbf{0} \\ \hline \mathbf{0} & c\,B_d\,\boldsymbol{T}(0,c,d,e) \end{array}\right] && \text{(4.18b)} \\
&=\; T_e \left[\begin{array}{c|c} 1 & \mathbf{0} \\ \hline \mathbf{0} & c\,B_d \end{array}\right] \left[\begin{array}{c|c} 1 & \mathbf{0} \\ \hline \mathbf{0} & \boldsymbol{T}(0,c,d,e) \end{array}\right] . && \text{(4.18c)}
\end{aligned}$$

□

We now recall [4, Lemma 2.2(i)] which we require for our second proof.

Lemma 4.7 ([4, Lemma 2.2(i)]). *The matrix* $\boldsymbol{T} = \boldsymbol{T}(0, c, d, e)$ *satisfies the recurrence*

$$T(n,k) \;=\; e\,T(n-1,k) \;+\; \sum_{m=0}^{n-1} \binom{n-1}{m} d^m c\, T(n-1-m,k-1) \tag{4.19}$$

for $n \geq 1$*, where* $T(n,k) \stackrel{\text{def}}{=} 0$ *if* $n < 0$ *or* $k < 0$*.*

Second Proof of Proposition 4.1. We rewrite the recurrence (4.19) as

$$T(n,k) \;-\; e\,T(n-1,k) \;=\; c \sum_{m=0}^{n-1} \binom{n-1}{n-1-m} d^m\, T(n-1-m,k-1)\;, \tag{4.20}$$

which in matrix form reads

$$L_{-e}\, \boldsymbol{T}(0,c,d,e) \;=\; \left[\begin{array}{c|c} 1 & \mathbf{0} \\ \hline \mathbf{0} & c\,B_d \end{array}\right] \left[\begin{array}{c|c} 1 & \mathbf{0} \\ \hline \mathbf{0} & \boldsymbol{T}(0,c,d,e) \end{array}\right] . \tag{4.21}$$

Left-multiplying both sides by $(L_{-e})^{-1} = T_e$ yields (4.5). □

4.2 A conjectural generalisation of the real-rootedness of the Eulerian polynomials

For $n \geq 0$, let $T_n(x) = \sum_{k=0}^{n} T(n,k)x^k$ denote the generating polynomial of the n-th row of the matrix $\boldsymbol{T}$ defined by the linear recurrence (4.1). We have the following recurrence:

Proposition 4.8. *The polynomials* $T_n(x) = \sum_{k=0}^{n} T(n,k)x^k$ *where the coefficients are given by* (4.1)*, satisfy the following recurrence relation:*

$$T_n(x) \;=\; \Big(\big(a(n-1)+c\big)\,x + e\Big)\, T_{n-1}(x) + x\,(d-ax)\,T'_{n-1}(x) + \big(f(n-2)+g\big)\,x\,T_{n-2}(x)\;. \tag{4.22}$$

The proof of Proposition 4.8 is straightforward and we omit the details. However, this recurrence gives us the following more interesting consequence.

Proposition 4.9. *Let* a, c, d, e, f, g *be non-negative real-numbers. Then for all* $n \geq 0$*, the polynomial* $T_n(x)$ *is real-rooted.*

PROOF. Define polynomials $p(x), q(x)$ by $p(x) = x(d-ax)$ and $q_n(x) = (f(n-2)+g)x$. For every $x \le 0$, we always have $p(x) \le 0$ and $q(x) \le 0$. The proposition then follows from [15, Corollary 2.4]. □

Proposition 4.9 naturally suggests the following upgrade:

Conjecture 4.10. *Let a, c, d, e, f, g be indeterminates. For every fixed $n \ge 0$, the Toeplitz matrix $T_\infty((T(n,k))_{k\ge 0})$ is coefficientwise totally positive in all six indeterminates a, c, d, e, f, g.*

Notice that the two cases $a = f = g = 0$ and $d = f = g = 0$ in Conjecture 4.10 are covered by Theorem 4.3.

5 Application: Ordinary and Exponential Riordan arrays

DEFINE RIORDAN ARRAYS!!!!!

Proposition 5.1. *(a) Let $F = \sum_{n=0}^{\infty} f_n t^n$ and let $G = \sum_{n=0}^{\infty} g_n t^n$ be formal power series with $g_0 = 0$, and $f_0 \neq 0$ is invertible. The ordinary Riordan array $\mathcal{R}(F, G)$ has the following matrix as its left production matrix:*

$$Q \;=\; \mathcal{R}(F,t) \,\cdot\, \left[\begin{array}{c|c} 1 & \\ \hline & \mathcal{R}(G/(tF),t) \end{array}\right] . \tag{5.1}$$

(b) Let $F = \sum_{n=0}^{\infty} f_n t^n/n!$ and let $G = \sum_{n=0}^{\infty} g_n t^n/n!$ be formal power series with $g_0 = 0$, and $f_0 \neq 0$ is invertible. The exponential Riordan array $\mathcal{R}[F, G]$ has the following matrix as its left production matrix:

$$Q \;=\; \mathcal{R}[F,t] \,\cdot\, \left[\begin{array}{c|c} 1 & \\ \hline & \mathcal{R}[G'/F,t] \end{array}\right] . \tag{5.2}$$

PROOF. We only exhibit our computation for (a).

$$\begin{aligned} Q \;&=\; \mathcal{R}(F,G) \,\cdot\, \left[\begin{array}{c|c} 1 & \\ \hline & \mathcal{R}(F,G)^{-1} \end{array}\right] \\ &=\; \mathcal{R}(F,t) \,\cdot\, \mathcal{R}(1,G) \,\cdot\, \left[\begin{array}{c|c} 1 & \\ \hline & \mathcal{R}(1,\overline{G}) \,\cdot\, \mathcal{R}(1/F,t) \end{array}\right] \end{aligned} \tag{5.3}$$

The rest follows by writing $\mathcal{R}(1,G) = \left[\begin{array}{c|c} 1 & \\ \hline & \mathcal{R}(G/t,G) \end{array}\right]$ and then multiplying the three terms using the binary operation of the Riordan group.

The proof of (b) is similar and we omit the details. □

Remarks. Let us consider the situation when the exponential Riordan array $\mathcal{R}[F, G]$ belongs to the *derivative subgroup*, i.e., $F = G'$. In this case, as $\mathcal{R}[1, t]$

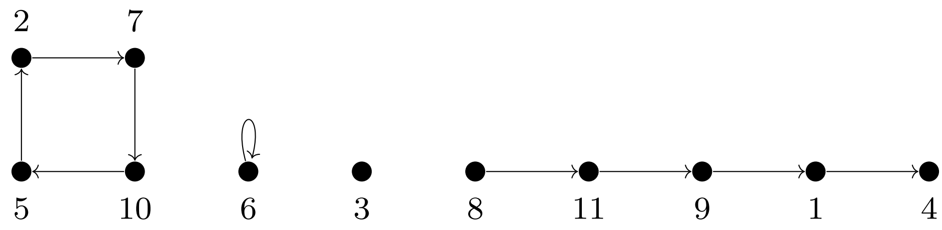

Figure 1: A Laguerre digraph on 11 vertices with 2 cycles (one of which is a loop) and 2 paths (one of which is an isolated vertex).

is the identity matrix, we get that the left production matrix of $\mathcal{R}[G', G]$ is simply $Q = \mathcal{R}[F, t]$; this was noticed in [5, Proof of Theorem 4.2].

Analogously, for ordinary Riordan arrays, when $\mathcal{R}(F, G)$ belongs to the *Bell subgroup*, i.e., $G = tF$, we get that the left production matrix of $\mathcal{R}(F, tF)$ is simply $Q = \mathcal{R}(F, t)$.

Notice that in both situations, the left production matrix is itself a (exponential or ordinary, resp.) Riordan array that belongs to the *Appell subgroup*. $\blacksquare$

5.1 Application: Multivariate Laguerre coefficient matrix

USE THE EXPONENTIAL RIORDAN ARRAY STRUCTURE OF LAGUERRE COEFFICIENT MATRIX AND WRITE THIS OUT [7]!!!!

A ***Laguerre digraph*** is a directed graph in which each vertex has out-degree 0 or 1 and in-degree 0 or 1. It follows that each weakly connected component of a Laguerre digraph is either a directed path of some length $\ell \geq 0$ (where a path of length 0 is an isolated vertex) or else a directed cycle of some length $\ell \geq 1$ (where a cycle of length 1 is a loop): see Figure 1. Laguerre digraphs were introduced by Foata and Strehl [11] as Laguerre configurations; however the definition that we use here was first defined by Sokal in [19].

For each integer $n \geq 0$, let us write $\mathbf{LD}_n$ for the set of Laguerre digraphs on the vertex set $[n] \stackrel{\text{def}}{=} \{1, \ldots, n\}$; and for $n \geq k \geq 0$, let us write $\mathbf{LD}_{n,k}$ for the set of Laguerre digraphs on the vertex set $[n]$ with k paths. For $G \in \mathbf{LD}_n$, we write $\mathrm{e}(G)$ [resp. $\mathrm{cyc}(G)$, $\mathrm{pa}(G)$] for the number of edges (resp. cycles, paths) in G, and observe that $\mathrm{e}(G) = n - \mathrm{pa}(G)$.

In Section 5.1.1, we follow [7] and recall several statistics on Laguerre digraphs and define the multivariate Laguerre coefficient matrix and the multivariate Laguerre polynomials. Then in Section 5.1.2, we compute the left production matrix of the multivariate Laguerre coefficient matrix.

5.1.1 Statistics on Laguerre digraphs and the multivariate Laguerre coefficientwise matrix

Throughout this section, we make the following convention about *boundary conditions* at the two ends of a path. We will here use 0–0 boundary conditions: that is, we extend the Laguerre digraph G on the vertex set $[n]$ to a digraph $\widehat{G}$ on the vertex set $[n] \cup \{0\}$ by decreeing that any vertex $i \in [n]$ that has in-degree (resp. out-degree) 0 in G will receive an incoming (resp. outgoing) edge from (resp. to) the vertex 0. In this way each vertex $i \in [n]$ will have a unique predecessor $p(i) \in [n] \cup \{0\}$ and a unique successor $s(i) \in [n] \cup \{0\}$. We then say that a vertex $i \in [n]$ is a

- *peak* (p) if $p(i) < i > s(i)$;
- *valley* (v) if $p(i) > i < s(i)$;
- *double ascent* (da) if $p(i) < i < s(i)$;
- *double descent* (dd) if $p(i) > i > s(i)$;
- *fixed point* (fp) if $p(i) = i = s(i)$.

(Note that "fixed point" is a synonym of "loop".) When these concepts are applied to the cycles of a Laguerre digraph, we obtain the usual ***cycle classification*** of indices in a permutation as cycle peaks, cycle valleys, cycle double rises, cycle double falls and fixed points [21,26]. When applied to the paths of a Laguerre digraph, we obtain the usual ***linear classification*** of indices in a permutation (written in word form) as peaks, valleys, double ascents or double descents [22, p. 45]. Note that, because of the 0–0 boundary conditions, an isolated vertex is always a peak, the initial vertex of a path is always a peak or double ascent, and the final vertex of a path is always a peak or double descent; moreover, each path contains at least one peak.

We write $\mathrm{p}(G), \mathrm{val}(G), \mathrm{da}(G), \mathrm{dd}(G), \mathrm{fp}(G)$ for the number of vertices $i \in [n]$ that are, respectively, peaks, valleys, double ascents, double descents or fixed points. We then introduce the ***multivariate Laguerre coefficient matrix***

$$\widehat{\mathsf{L}}^{(\lambda)}(y_{\mathrm{p}}, y_{\mathrm{val}}, y_{\mathrm{da}}, y_{\mathrm{dd}}, y_{\mathrm{fp}})_{n,k} \stackrel{\mathrm{def}}{=} \sum_{G \in \mathbf{LD}_{n,k}} y_{\mathrm{p}}^{\mathrm{p}(G)} y_{\mathrm{v}}^{\mathrm{val}(G)} y_{\mathrm{da}}^{\mathrm{da}(G)} y_{\mathrm{dd}}^{\mathrm{dd}(G)} y_{\mathrm{fp}}^{\mathrm{fp}(G)} \, \lambda^{\mathrm{cyc}(G)} \,. \tag{5.4}$$

This polynomial is homogeneous of degree n in $y_{\mathrm{p}}, y_{\mathrm{val}}, y_{\mathrm{da}}, y_{\mathrm{dd}}, y_{\mathrm{fp}}$. The bivariate exponential generating function for this matrix was computed in [7, sec. 4.1].

Because each path contains at least one peak, we can, if we wish, remove a factor y_{p}^k and define a unit-lower-triangular matrix by

$$\widehat{\mathsf{L}}^{(\lambda)\flat}(y_{\mathrm{p}}, y_{\mathrm{val}}, y_{\mathrm{da}}, y_{\mathrm{dd}}, y_{\mathrm{fp}})_{n,k} \stackrel{\mathrm{def}}{=} \widehat{\mathsf{L}}^{(\lambda)}(y_{\mathrm{p}}, y_{\mathrm{val}}, y_{\mathrm{da}}, y_{\mathrm{dd}}, y_{\mathrm{fp}})_{n,k} / y_{\mathrm{p}}^k \,. \tag{5.5}$$

This polynomial is homogeneous of degree $n-k$ in $y_{\mathrm{p}}, y_{\mathrm{val}}, y_{\mathrm{da}}, y_{\mathrm{dd}}, y_{\mathrm{fp}}$.

Finally, we introduce the row-generating polynomials:

$$\widehat{\mathcal{L}}_n^{(\lambda)}(x; y_{\mathrm{p}}, y_{\mathrm{val}}, y_{\mathrm{da}}, y_{\mathrm{dd}}, y_{\mathrm{fp}}) \stackrel{\mathrm{def}}{=} \sum_{k=0}^{n} \mathsf{L}^{(\lambda)}(y_{\mathrm{p}}, y_{\mathrm{val}}, y_{\mathrm{da}}, y_{\mathrm{dd}}, y_{\mathrm{fp}})_{n,k} \, x^k \tag{5.6}$$

We refer to these as the ***multivariate Laguerre polynomials***.

However, we can go one step farther and assign different weights for the vertices belonging to a cycle or to a path. For a Laguerre digraph G, let us write $\mathrm{pcyc}(G)$, vcyc(G), dacyc(G), ddcyc(G), fp(G) for the number of peaks, valleys, double ascents, double descents and fixed points that belong to a cycle of G, and ppa(G), vpa(G), dapa(G), ddpa(G) for the number of peaks, valleys, double ascents and double descents that belong to a path of G (of course fixed points can only belong to a cycle). We then assign weights $y_{\rm p}, y_{\rm val}, y_{\rm da}, y_{\rm dd}, y_{\rm fp}$ to the vertices belonging to a cycle, and weights $z_{\rm p}, z_{\rm val}, z_{\rm da}, z_{\rm dd}$ to the vertices belonging to a path. We therefore define the ***generalized second multivariate Laguerre coefficient matrix***

$$\begin{aligned}
&\widetilde{\mathsf{L}}^{(\lambda)}(y_{\rm p}, y_{\rm val}, y_{\rm da}, y_{\rm dd}, y_{\rm fp}, z_{\rm p}, z_{\rm val}, z_{\rm da}, z_{\rm dd})_{n,k} \\
&\quad \stackrel{\rm def}{=} \sum_{G \in \mathbf{LD}_{n,k}} y_{\rm p}^{\rm pcyc(G)} y_{\rm v}^{\rm vcyc(G)} y_{\rm da}^{\rm dacyc(G)} y_{\rm dd}^{\rm ddcyc(G)} y_{\rm fp}^{\rm fp(G)}\, z_{\rm p}^{\rm ppa(G)} z_{\rm v}^{\rm vpa(G)} z_{\rm da}^{\rm dapa(G)} z_{\rm dd}^{\rm ddpa(G)} \;\times \\
&\qquad\qquad \lambda^{\rm cyc(G)}\,. \qquad (5.7)
\end{aligned}$$

This polynomial is homogeneous of degree n in $y_{\rm p}, y_{\rm val}, y_{\rm da}, y_{\rm dd}, y_{\rm fp}, z_{\rm p}, z_{\rm val}, z_{\rm da}, z_{\rm dd}$.

Because of the 0–0 boundary conditions, each path contains at least one peak; so we can, if we wish, remove a factor $z_{\rm p}^k$ and define a unit-lower-triangular matrix by

$$\widetilde{\mathsf{L}}^{(\alpha)\flat}(y_{\rm p}, y_{\rm val}, y_{\rm da}, y_{\rm dd}, y_{\rm fp}, z_{\rm p}, z_{\rm val}, z_{\rm da}, z_{\rm dd})_{n,k} \;\stackrel{\rm def}{=}\; \widetilde{\mathsf{L}}^{(\alpha)}(y_{\rm p}, y_{\rm val}, y_{\rm da}, y_{\rm dd}, y_{\rm fp}, z_{\rm p}, z_{\rm val}, z_{\rm da}, z_{\rm dd})_{n,k}/z_{\rm p}^k\,. \qquad (5.8)$$

This polynomial is homogeneous of degree $n-k$ in $y_{\rm p}, y_{\rm val}, y_{\rm da}, y_{\rm dd}, y_{\rm fp}, z_{\rm p}, z_{\rm val}, z_{\rm da}, z_{\rm dd}$.

5.1.2 Left production matrices for the multivariate Laguerre coefficient matrix

We now compute the left production matrix for the generalized second multivariate Laguerre coefficient matrix.

We first make the following two simple observations which follow from the definition:

$$\widetilde{\mathsf{L}}^{(0)}(y_{\rm p}, y_{\rm val}, y_{\rm da}, y_{\rm dd}, y_{\rm fp}, z_{\rm p}, z_{\rm val}, z_{\rm da}, z_{\rm dd})_{n,1} \;=\; \widetilde{\mathsf{L}}^{(0)}(0,0,0,0,0, z_{\rm p}, z_{\rm val}, z_{\rm da}, z_{\rm dd})_{n,1} \qquad (5.9a)$$

$$\widetilde{\mathsf{L}}^{(\lambda)}(y_{\rm p}, y_{\rm val}, y_{\rm da}, y_{\rm dd}, y_{\rm fp}, z_{\rm p}, z_{\rm val}, z_{\rm da}, z_{\rm dd})_{n,0} \;=\; \widetilde{\mathsf{L}}^{(\lambda)}(y_{\rm p}, y_{\rm val}, y_{\rm da}, y_{\rm dd}, y_{\rm fp}, 0,0,0,0)_{n,0} \qquad (5.9b)$$

Proposition 5.2 (Left production matrix for the multivariate Laguerre coefficient matrix). *The matrix $\widetilde{\mathsf{L}}^{(\lambda)}$ has the left production matrix $Q = (q_{nk})_{n,k\ge 0}$ whose entries are given for $k=0$ and $n \ge 0$ as*

$$q_{n0} \;=\; \widetilde{\mathsf{L}}^{(\lambda)}(y_{\rm p}, y_{\rm val}, y_{\rm da}, y_{\rm dd}, y_{\rm fp}, 0,0,0,0)_{n,0} \qquad (5.10)$$

and for $n \ge k \ge 1$

$$\begin{aligned}
q_{nk} \;=\; \binom{n-1}{k-1} \Big(& \widetilde{\mathsf{L}}^{(0)}(0,0,0,0,0, z_{\rm p}, z_{\rm val}, z_{\rm da}, z_{\rm dd})_{n-(k-1),1} \\
& + \lambda\, \Big([\lambda^1] \widetilde{\mathsf{L}}^{(\lambda)}(y_{\rm p}, y_{\rm val}, y_{\rm da}, y_{\rm dd}, y_{\rm fp}, 0,0,0,0)_{n-(k-1),0} \Big) \Big)\,. \qquad (5.11)
\end{aligned}$$

WARNING!!!!!! THIS IS INCORRECT AND DOES NOT MATCH COMPUTATIONS EVEN WHEN ALL STATISTICS ARE SET TO 1 EXCEPT FOR LAMBDA!!!!!! I CANNOT FIGURE OUT WHAT IS INCORRECT IN MY WRONG PROOF HERE!!!!!!

PROOF. First consider a Laguerre digraph $G \in \mathbf{LD}_{n+1,k+1}$ and let us look at the connected component containing the vertex $n+1$. Let us use G' to denote the graph obtained by removing this component and let i be the number of vertices of G'; here G' has k components and $0 \le i \le n$.

On the other hand, let us instead start with a graph $G'' \in \mathbf{LD}_{i,k}$ and count the number of ways to insert a new connected component with possibly relabeling the vertices but maintaining their total order, such that this new component contains the vertex $n+1$. In this new graph G, labels of the subgraph corresponding to G'' can be chosen in $\binom{n}{i}$ ways (not including $n+1$). We have two choices for the new component:

- We insert a new path. In this case, we think of the connected component as a Laguerre digraph with a single path on the remaining $n+1-i$ labels. The weighted number of possibilities here is $\widetilde{\mathsf{L}}^{(0)}_{n+1-i,1}$.
- We insert a new cycle. In this case, we think of the connected component as a Laguerre digraph with a single cycle (and no paths) on the remaining $n+1-i$ labels. The weighted number of possibilities here is $[\lambda^1]\widetilde{\mathsf{L}}^{(\lambda)}_{n+1-i,0}$.

Putting all of this together gives us

$$\widetilde{\mathsf{L}}^{(\lambda)}_{n+1,k+1} \;=\; \sum_i \binom{n}{i} \left(\widetilde{\mathsf{L}}^{(0)}_{n+1-i,1} \,+\, \lambda\, \left([\lambda^1]\widetilde{\mathsf{L}}^{(\lambda)}_{n+1-i,0} \right) \right) \widetilde{\mathsf{L}}^{(\lambda)}_{i,k}\;. \tag{5.12}$$

Finally we conclude the proof by mentioning that value of q_{n0} in (5.10) is exactly as required by (2.3b)

□

Next, we factorise the left production matrix obtained in Proposition 5.2. Let D be the diagonal matrix whose 00-entry is 1 and for $n \ge 1$ its diagonal entries are given by the sequence $((n-1)!)_{n\ge 1}$. Also let $\widetilde{Q} = (\widetilde{q}_{nk})$ whose entries are given by

$$\widetilde{q}_{nk} \;=\; \begin{cases} \widetilde{\mathsf{L}}^{(\lambda)}_{n,0} & \text{if } n \ge k = 0 \\[2mm] \dfrac{\widetilde{\mathsf{L}}^{(0)}_{n-(k-1),1} \,+\, \lambda\,[\lambda^1]\widetilde{\mathsf{L}}^{(\lambda)}_{n-(k-1),0}}{(n-k)!} & \text{if } n \ge k \ge 1 \\[2mm] 0 & \text{otherwise} \end{cases}\;. \tag{5.13}$$

Here it is straightforward to see that Q can be factorised as follows:

$$Q \;=\; D\widetilde{Q}D \tag{5.14}$$

NOW USE THE EGF OF [26] AND FIND THE SPECIALISATIONS THAT MAKES THE LEFT PROD MAT TP!!!!!!!!!

6 Application: Linearly ordered phylogenetic trees

In [8, Final Remark in Section 4.3], the authors *phylogenetic trees* with linearly ordered children, and formulated [8, Conjectures 4.17]. We will prove [8, Conjectures 4.17(a)(b)] in this section.

We first recall the relevant combinatorial definitions. A ***phylogenetic tree*** is a leaf-labeled rooted tree (i.e. the leaves are labeled and the non-leaf vertices are unlabeled) in which all non-leaf vertices have at least two children. A ***linearly ordered phylogenetic tree*** is a phylogenetic tree such that the children at each vertex are linearly ordered. Let $\mathcal{D}_{n,k}$ denote the set of all linearly ordered phylogenetic trees with $n+1$ leaves and k internal (i.e. non-leaf) vertices; let $\widehat{\left\{ {n \atop k} \right\}}$ denote the cardinality of this set $\widehat{\left\{ {n \atop k} \right\}} := |\mathcal{D}_{n,k}|$. This number has a closed form expression as was observed in [8, Proposition 4.16, eq. (4.39)]:

$$|\mathcal{D}_{n,k}| \;=\; \widehat{\left\{ {n \atop k} \right\}} \;=\; \frac{(n+k)!}{k!}\binom{n-1}{k-1}\;; \tag{6.1}$$

these numbers match [16, A357367]. Let $d_n(x)$ denote the generating polynomials

$$d_n(x) \;\stackrel{\text{def}}{=}\; \sum_{k=0}^{n} \widehat{\left\{ {n \atop k} \right\}} x^k\;, \tag{6.2}$$

and let D be the lower-triangular matrix $D = \left(\widehat{\left\{ {n \atop k} \right\}} \right)_{n,k\ge 0}$.

It was conjectured [8, Conjecture 4.17] that the matrices D, $\overleftarrow{D}$ are totally positive, the sequence $(d_n(x))_{n\ge0}$ is coefficientwise Hankel totally positive, and it was proved that the polynomials $d_n(x)$ are negative-real-rooted [8, Theorem 4.18]. Here we will now prove that the matrices D, $\overleftarrow{D}$ are both totally positive and provide two proofs. Our first proof involves a matrix factorization for both matrices. For our second proof, we will compute the left production matrix of D and show that it is totally positive; then by virtue of Theorem 2.3 we obtain that the matrices D, $\overleftarrow{D}$ are both totally positive and also a new proof of the real-rootedness of the polynomials $d_n(x)$.

Theorem 6.1 (Total positivity for linearly ordered phylogenetic trees)**.** *The matrix* $D = \left(\widehat{\left\{ {n \atop k} \right\}} \right)_{n,k\ge 0}$ *and its reverse* $\overleftarrow{D}$ *are both totally positive.*

First Proof of Theorem 6.1. We first show that D is totally positive. First let $M = \left(\binom{n+k}{n-k} \right)_{n,k\ge 0}$ be the Morgan–Voyce matrix which is known to be totally positive as follows: the matrix M is the (ordinary) Riordan array

$$M \;=\; \mathcal{R}(1/(1-t), t/(1-t)^2) \tag{6.3}$$

and as the coefficients of $1/(1-t), t/(1-t)^2$ both form a Polya frequency sequence by using [6, Theorem 2.1], we get that M is totally positive (see [6] for details).

Next, let D_1, D_2 be the diagonal matrices whose 00-entries are 1 and the other diagonal entries are given by the sequences $\big((n-1)!\big)_{n\geq 1}$, $\big(\frac{(2k)!}{k!(k-1)!}\big)_{k\geq 1}$. We now notice that

$$\widehat{\left\{ n \atop k \right\}} = (n-1)! \cdot \binom{n+k}{n-k} \cdot \frac{(2k)!}{k!(k-1)!} \,. \tag{6.4}$$

which gives us the matrix factorisation $D = D_1 \cdot M \cdot D_2$. As each of the constituent matrices in this product are totally positive, the matrix D is totally positive.

Next, we show that $\overleftarrow{D}$ is totally positive. Here, let $N = \big(\frac{1}{k+1}\binom{n+1}{k}\binom{n}{k}\big)_{n,k\geq 0}$ be the matrix of Narayana numbers (of type A), and let $B = \big(\binom{n}{k}\big)_{n,k\geq 0}$ be the triangle of binomial coefficients. Finally, let D_3 be the diagonal matrix with diagonal entries given by the sequence $((n+2)!)_{n\geq 0}$. Here the matrix $\overleftarrow{D}$ has the following matrix factorisation:

$$\overleftarrow{D} \;=\; \left[\begin{array}{c|c} 1 & \mathbf{0} \\ \hline \multicolumn{2}{c}{D_3 \cdot N \cdot B} \end{array}\right] \;; \tag{6.5}$$

it is straightforward to check this factorisation and we omit the details. In [25], Wang and Yang showed that the Narayana triangle N is totally positive. It is well known that the binomial matrix B and also the matrix D_3 is clearly totally positive. All of this together implies that the matrix $\overleftarrow{D}$ is totally positive. $\square$

We will now state our second proof which is more relevant to the contents of this paper. More precisely, we will establish the following left production matrix:

Proposition 6.2. *(a) The matrix D enumerating linearly ordered phylogenetic trees has the left production matrix $Q = (q_{nk})_{n,k\geq 0}$ whose entries q_{nk} are given by*

$$q_{nk} \;=\; \begin{cases} 1 & \textit{if } n = k = 0 \\ \dfrac{(n-1)!}{k!}\,(2k-1)\,(n+k)(n-k+1) & \textit{if } n \geq k > 0 \\ 0 & \textit{otherwise} \end{cases} \;. \tag{6.6}$$

(b) The matrix Q is totally positive.

Proof of Proposition 6.2(a). We will show that the numbers $\widehat{\left\{ n \atop k \right\}}$ satisfy the following recurrence relation:

$$\widehat{\left\{ n+1 \atop k+1 \right\}} \;=\; \sum_{i=0}^{n} q_{n+1,i+1} \widehat{\left\{ i \atop k \right\}} \;=\; \sum_{i=k}^{n} \frac{n!}{(i+1)!}\,(2i{+}1)\,(n{+}i{+}2)(n{-}i{+}1)\,\widehat{\left\{ i \atop k \right\}} \,. \tag{6.7}$$

[2] Consider a tree $T \in \mathcal{D}_{n+1,k+1}$. **I AM STUCK NOW!!!! UGH!!!! I COULD**

---

[2]**Remark.** We remark that these numbers also satisfy the following recurrence:

$$\widehat{\left\{ n+1 \atop k+1 \right\}} \;=\; \sum_{i=k}^{n} \binom{n+k+1}{n+1-i} (n+2-i)! \widehat{\left\{ i \atop k \right\}} \,, \tag{6.8}$$

which is interesting because the coefficients multiplying the numbers $\widehat{\left\{ i \atop k \right\}}$ are different from those in (6.7).

USE WZ BUT I WILL GET NO COMBINATORIAL INSIGHT IF I DO THAT!!!! NEED BIJECTIVE PROOF OF THIS!!!!
□

PROOF OF PROPOSITION 6.2(B). Let D_1, D_2, D_3 be the diagonal matrices whose diagonal entries are the three sequences $((n)!)_{n\geq 0}, (2n+2)_{n\geq 0}, ((2k+1)/(k+1)!)_{k\geq 0}$, respectively. Also, let T denote the lower triangular matrix all of whose non-zero entries are 1. We claim that the left production matrix Q has the following matrix factorisation:

$$Q \;=\; \left[\begin{array}{c|c} 1 & \mathbf{0} \\ \hline \mathbf{0} & D_1\, T\, D_2\, T\, D_3 \end{array}\right] . \tag{6.9}$$

This is straightforward, we compute the entries and then compare the resulting expression with (6.6). As each of the constituent matrices D_1, D_2, D_3, T are totally positive, hence the matrix Q is also totally positive. □

SECOND PROOF OF THEOREM 6.1. The theorem follows by combining Proposition 6.2 with Theorem 2.3(a),(b). □

A further consequence of Proposition 6.2 combined with Theorem 2.3(c) is a new proof of the following fact that was previously observed in [8, Theorem 4.18]:

Corollary 6.3. *For every $n \geq 1$, the polynomials $d_n(x)$ are negative real-rooted.*

Remark. The matrix $\Delta \overleftarrow{D}$ is lower triangular with non-zero diagonal entries. We remark that its left production matrix is not even TP_1. ■

We finish this section by stating another combinatorial interpretation for the numbers $\widehat{\left\{ {n \atop k} \right\}}$ that was observed in [24].

Proposition 6.4. *The number $\widehat{\left\{ {n \atop k} \right\}}$ counts the number of ordered set partitions of $n+k$ elements into k blocks, where each block is of length greater than or equal to 2, such that the elements in each block have a total order among them.*

WRITE THE PROOF WHICH IS VIA MULTIVARIATE WARD POLYNOMIALS!!!!

7 Matrices which satisfy a subset of the four positivities

Let $A = (a_{n,k})_{n,k\geq 0}$ be a lower triangular matrix. We asked four different questions and it is natural to ask if there is any relation between these questions. However, the answer is no. For a given matrix A, the answer to these four questions can be simultaneously be one of the 16 quadruples of True/False. We will now exhibit a matrix A for each of these 16 quadruples and thus show that, in general, there are no relations between these four questions. In fact, taking the symmetry of A and $\overleftarrow{A}$ into consideration we only exhibit the following 11 examples.

(a) (T,T,T,T): We have already mentioned several examples of such matrices; these include the binomial triangle $\left(\binom{n}{k}\right)_{n,k\ge 0}$, the Stirling subset triangle $\left(\left[{n \atop k}\right]\right)_{n,k\ge 0}$, the Stirling cycle triangle $\left(\left\{{n \atop k}\right\}\right)_{n,k\ge 0}$.

(b) (T,F,T,T)

(c) (T,T,F,T)

(d) (T,T,T,F): Delannoy triangle

(e) (F,F,T,T)

(f) (T,F,F,T): Ballot table

(g) (T,F,T,F): Bidiagonal matrix, Motzkin matrix

(h) (T,T,F,F): All ones matrix

(i) (T,F,F,F): Consider the following matrix A:

$$\widehat{A} \;=\; \begin{bmatrix} 1 & & & \\ 1 & 1 & & \\ 1 & 2 & 1 & \\ 0 & 1 & 1 & 1 \end{bmatrix} \tag{7.1}$$

and construct $A = \left[\begin{array}{c|c} \widehat{A} & \mathbf{0} \\ \hline \mathbf{0} & \mathbf{0} \end{array}\right]$

(j) (F,F,T,F): Consider the following matrix A:

$$\widehat{A} \;=\; \begin{bmatrix} 1 & & & \\ 1 & 1 & & \\ 1 & 100 & 1 & \\ 32 & 32 & 10 & 1 \end{bmatrix} \tag{7.2}$$

and construct $A = \left[\begin{array}{c|c} \widehat{A} & \mathbf{0} \\ \hline \mathbf{0} & \mathbf{0} \end{array}\right]$

(k) (F,F,F,T)

Acknowledgments

We wish to thank Sylvie Corteel, Sergey Fomin, Yi Wang, Sai-Nan Zheng and ?????? for helpful correspondence.

This research was supported in part by the U.K. Engineering and Physical Sciences Research Council grant EP/N025636/1, a fellowship from the China Scholarship Council, and a fellowship from the Deutsche Forschungsgemeinschaft. The second author is currently supported by a Tsinghua University Shuimu scholarship.

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